

AutoBio+: Benchmarking Dexterous Manipulation with Dynamic Experimental Phenomena in Realistic Laboratories

Anonymous Authors



Fig. 1: **Overview of AutoBio+.** AutoBio+ is a benchmark for dexterous manipulation in biological laboratories, comprising 14 tasks across four robot embodiments and incorporating three dynamic experimental phenomena, including flames, vapor, and bubbles. Its hybrid rendering pipeline integrates interactive simulated foregrounds with 3D Gaussian Splatting (3DGS) [1] backgrounds reconstructed from six real laboratory scenes, while parallel simulation supports efficient data generation and evaluation.

Abstract—Automating biological laboratories requires robots to perform dexterous tool interactions under realistic visual conditions while handling dynamic experimental phenomena, presenting significant challenges for robotic manipulation. To enable systematic evaluation under these conditions, we introduce AutoBio+, a simulation benchmark and framework for dexterous manipulation in realistic biological laboratory environments. AutoBio+ comprises 14 tasks across 4 robot embodiments, spanning functional tool use, precision manipulation, language-conditioned visual grounding, and instrument operation. We introduce controllable dynamic experimental phenomena, including flames, vapor, and bubbles, to capture the evolving visual conditions of laboratory procedures. To further enhance environmental realism, we develop a hybrid rendering pipeline that integrates interactive simulated foreground objects with Gaussian-splatting backgrounds reconstructed from 6 real laboratory scenes. For efficient large-scale data generation and evaluation, we further build a parallel simulation framework on

top of mjlab, proposing Parallel Finite-State Machine (PFSM) and integrating batched rendering with decoupled trajectory replay for demonstration generation and model-independent closed-loop policy evaluation. Our evaluation reveals substantial variation in policy performance across tasks, exposing distinct strengths and weaknesses of current manipulation policies. Real-world experiments further demonstrate a reduced sim-to-real gap, supporting the validity of AutoBio+ for realistic manipulation evaluation. Supplementary materials are available at <https://autobio-plus.github.io/AutoBioPlus>.

I. INTRODUCTION

Automated laboratories, often referred to as AutoLabs, are transforming scientific experimentation by shifting routine procedures from manual operation toward autonomous execution [9], [10]. Conventional laboratory automation, how-

TABLE I: Comparison of AutoBio+ with existing benchmarks. ✓ denotes full support, ✓ denotes partial support, and ✗ denotes no support.

Benchmark	Biolab	Dexterous Manipulation	Realistic Rendering	Parallelization	Multi-Embodiment	Dynamic Phenomena
+ Bi-DexHands [2]	✗	✓	✗	✓	✗	✗
DexVerse [3]	✗	✓	✓	✓	✓	✗
DexJoCo [4]	✗	✓	✗	✗	✗	✗
Chemistry3D [5]	✓	✗	✗	✓	✗	✓
LabUtopia [6]	✓	✗	✗	✗	✗	✗
Labimus [7]	✓	✓	✓	✗	✗	✗
AutoBio [8]	✓	✓	✓	✓	✓	✗
AutoBio+ (Ours)	✓	✓	✓	✓	✓	✓

ever, typically relies on task-specific instruments and tightly coupled workflows. Although such systems are efficient in structured settings, adapting them to new experimental protocols, tools, or operating conditions often requires substantial hardware modification and system reintegration. General-purpose robotic manipulators equipped with dexterous hands or grippers provide a more flexible alternative, allowing laboratory procedures to be adapted through demonstrations and learned policies rather than hardware-specific re-engineering. This flexibility is particularly important in biological laboratories, where robots must perform fine-grained operations such as manipulating pipettes and tweezers, inserting tubes, actuating instruments, and coordinating multiple objects over long-horizon procedures.

Recent robot-learning methods, particularly vision-language-action (VLA) policies, have demonstrated strong manipulation capabilities in household and industrial environments [11]–[13]. To evaluate these learned policies, recent benchmarks have explored manipulation from several complementary perspectives. General-purpose manipulation benchmarks such as RoboTwin 2.0 [14] and LIBERO [15] provide diverse household and industrial tasks with controlled object interactions. Scientific robotics benchmarks [5], [6], [8] introduce laboratory-specific objects and procedures, while dexterous manipulation benchmarks such as DexJoCo [4], DexVerse [3], and Bi-DexHands [2] focus on complex hand–object coordination. Recently, Labimus [7] extends dexterous manipulation evaluation to chemical laboratory scenarios. However, laboratory procedures often introduce temporally varying visual conditions, such as flames, vapor, and bubbles, which are underrepresented in existing manipulation benchmarks.

To address this gap, we introduce **AutoBio+**, a simulation benchmark for dexterous manipulation in realistic biological laboratory environments. AutoBio+ comprises 14 tasks across four robot embodiments, covering functional tool use, precision manipulation, language-conditioned visual grounding, and instrument operation. It incorporates controllable dynamic experimental phenomena, including flames, vapor, and bubbles, to capture visually evolving conditions during laboratory procedures. To improve visual realism while preserving physical interaction, AutoBio+ proposes a hybrid rendering strategy that combines interactive simulated foreground objects with Gaussian-splatting backgrounds reconstructed from six real laboratory scenes. Leveraging

mjlab as the simulation foundation, we design a parallel simulation framework centered around the Parallel Finite-State Machine (PFSM) policy, together with batched rendering and decoupled trajectory replay, enabling efficient demonstration generation and model-independent closed-loop evaluation.

Our contributions are summarized as follows:

- **Multi-Embodiment Benchmark for Dexterous Laboratory Manipulation.** We introduce AutoBio+, comprising 14 laboratory manipulation tasks across four robot embodiments and covering functional tool use, precision manipulation, language-conditioned visual grounding, and instrument operation.
- **Dynamic Experimental Phenomena.** We incorporate controllable flames, vapor, and bubbles into laboratory procedures, introducing process-dependent visual changes for evaluating learned policies under temporally varying and perceptually complex observations.
- **Photorealistic Hybrid Rendering.** We combine interactive MuJoCo-rendered foreground objects with Gaussian-splatting backgrounds reconstructed from real laboratories, improving visual realism while preserving physical interaction and efficient batched rendering.
- **Parallel Data Generation and Evaluation.** Built on mjlab, AutoBio+ supports parallel demonstration collection with the proposed PFSM policy, rendering, and closed-loop policy evaluation, enabling efficient large-scale data generation and benchmarking.

II. RELATED WORK

Visuomotor Policies. Visuomotor policies for robot manipulation have evolved from directly regressing actions from visual observations toward increasingly structured generative policies. Early Transformer-based approaches predict temporally coherent action chunks from visual observations, improving the consistency of long-horizon execution [11]. Diffusion-based imitation policies then model the action distribution through iterative denoising, providing a flexible way to represent multimodal behaviors and contact-rich motions [12]. Building on these low-level visual policies, VLA models incorporate large vision-language representations and language-conditioned action generation, enabling a single policy to connect scene understanding, task instructions, and continuous control [13]. As VLA research has advanced, it has become increasingly clear that predicting how the world will evolve under candidate actions can provide useful

temporal and physical information for manipulation. This insight has led to World Action Models (WAMs), which augment the VLA formulation by jointly learning action generation and future visual-state prediction [16], [17]. Despite these advances, existing visuomotor policies have been developed and evaluated predominantly on general-purpose manipulation tasks. Their capabilities in scientific laboratory settings, particularly for the fine-grained, contact-rich, and dexterous operations required in biological experiments, remain insufficiently explored. Our benchmark is designed to address the lack of evaluation for dexterous manipulation in photorealistic biological laboratory settings, featuring a diverse set of tasks that capture fine-grained operations.

Manipulation Benchmarks. Existing manipulation benchmarks commonly evaluate general robotic capabilities in household or industrial environments. Benchmarks such as RoboTwin 2.0 [14], RLBench [18], and BEHAVIOR-1K [19] cover a wide variety of tasks, and select parallel-jaw grippers as their robot embodiments. The low-dimensional control space of these grippers supports efficient trajectory collection and large-scale evaluation, yet it can hardly fully represent the fine-grained coordination required for human-like manipulation. In contrast, dexterous manipulation benchmarks such as Bi-DexHands [2], DexVerse [3], and DexJoCo [4] emphasize multi-fingered hand control and complex, contact-rich interactions. They provide valuable testbeds for studying coordinated finger motion, bimanual control, and generalization across objects, tasks, or robot embodiments. These benchmarks provide valuable foundations for studying dexterous manipulation, but laboratory-specific procedures involving diverse tools, instruments, and dynamic experimental phenomena remain less explored. Meanwhile, recent simulation frameworks have investigated the integration of neural rendering with interactive physics simulators to improve visual realism while preserving physical interaction [20]. However, such simulation advances have not yet been integrated into benchmarks for evaluating dexterous laboratory manipulation under realistic and evolving experimental conditions.

Scientific Robotics Benchmarks. Recent scientific robotics benchmarks have begun to investigate robotic manipulation in laboratory environments. Chemistry3D [5] and Lab-Utopia [6] provide scientific manipulation scenarios, but primarily focus on gripper-based interactions. AutoBio [8] extends this line of work to dexterous manipulation with a limited set of laboratory tasks, while simplifying hand control to binary open–close actions. In a related direction, Labimus [7] studies dexterous manipulation by humanoid robots in chemical laboratories. Against this backdrop, AutoBio+ broadens the scope of scientific robotics benchmarking with a larger and more diverse task suite, multiple robot embodiments, and richer dynamic visual effects, enabling a more comprehensive evaluation of robot-learning models.

III. AUTOBIO+ SIMULATION

To support realistic parallel evaluation in biological laboratory environments, we develop **AutoBio+ Simulation**, a photorealistic and parallelizable simulation framework built

on mjlab [21]. Our approach extends the existing mjlab infrastructure with enhanced rendering capabilities, dynamic experimental phenomena, and system-level parallelization. This design preserves the robustness and physical accuracy of the underlying physics engine while addressing its limitations in visual realism within biolab scenarios.

The framework comprises three key components: (A) a particle visual-effects system, (B) hybrid rendering, and (C) parallel simulation.

A. Particle System for Dynamic Experimental Phenomena

Biological laboratory environments often contain dynamic visual phenomena, such as visible vapor formation and time-varying flame intensity, which are perceptually important but expensive to model with full physics. To preserve scalability, we propose a **particle-based visual effect system** that favors perceptual fidelity while remaining lightweight and parallelizable. Each visual element is a particle primitive with state $\mathbf{s}_i(t) = (\mathbf{x}_i(t), \mathbf{c}_i(t), \alpha_i(t), \sigma_i(t))$. Here, $\mathbf{x}_i(t)$ denotes position, while $\mathbf{c}_i(t)$, $\alpha_i(t)$, and $\sigma_i(t)$ specify the particle color, opacity, and visual size, respectively. All particle states are decoupled from the scene physics and updated purely for rendering.

At each time step of duration Δt , particle positions are updated using a lightweight rule that combines deterministic drift with a stochastic perturbation:

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + [\mathbf{d}_i(\mathbf{x}_i, t) + \boldsymbol{\eta}_i(t)] \Delta t, \quad (1)$$

where $\mathbf{d}_i(\mathbf{x}_i, t)$ is a deterministic drift field that encodes effects such as buoyancy and upward motion, and $\boldsymbol{\eta}_i(t)$ is a small random displacement-rate perturbation that produces flicker and irregular motion. The scale $\sigma_i(t)$ may be varied over the particle lifetime to model expansion or growth. Particle opacity decays with age (e.g., $\alpha_i(t) = \alpha_{0,i} \exp(-(t - t_{0,i})/\tau_i)$, where τ_i is a predefined time constant), and particles are spawned from localized sources with controlled emission rates and parameter distributions, including initial scales. The design is inspired by real-time game-engine particle systems [22], producing perceptually convincing vapor, flame, and bubbles at low cost while remaining decoupled from the physics simulation. The effects are illustrated in the lower panel of Fig. 2.

Flame. The flame is approximated by upward-moving cube particles with temporally varying color and opacity. The particles follow stochastic upward drift, exhibit gradual alpha decay, and use color gradients to convey high-temperature cores and cooler edges. The flame color is also modulated by the rotational positions of the Bunsen burner’s two rings, causing its color to vary with burner adjustment.

Vapor. For phenomena such as the visible vapor produced by liquid nitrogen in a container, we model vapor as expanding particle clouds with randomized drift and dispersion. The spherical particles increase their spread over time, modulate their opacity based on particle age and local density, and respond to coarse airflow to indicate directional drift.

Bubbles. To represent bubble formation and stirring-induced motion, such as during heating or magnetic stirring, we

simulate rising particles with vortex-like trajectories and varying rise speeds. The particles combine upward buoyant motion with rotational offsets to approximate vortices and bubble paths.

B. Hybrid Rendering for Realistic Laboratory Environments

To achieve realistic laboratory environments while preserving physical interaction, we develop a hybrid rendering approach that integrates Gaussian-splatting backgrounds with simulator-rendered interactive foreground objects. Unlike replacing the simulator renderer entirely, our approach maintains the physical controllability of simulated objects while introducing realistic visual contexts. The 3DGS assets for six laboratory scenes are captured with a lightweight spatial scanner (Lixel K2)¹. The final image is formed as

$$I_{\text{final}} = M \odot I_{\text{origin}} + (1 - M) \odot \mathcal{R}(\{G_i\}_{i=1}^N, T_{\text{cam}}), \quad (2)$$

$$M \in \{0, 1\}^{H \times W}.$$

Here, M is mjlabs’ object-level foreground mask, I_{origin} is the original simulator image, $\{G_i\}_{i=1}^N$ are the preconstructed scene Gaussians, T_{cam} is the camera pose, $\mathcal{R}$ is the Gaussian splatting renderer, and $\odot$ denotes element-wise multiplication. This hybrid design retains interactive simulator objects while providing efficient, photorealistic laboratory backgrounds.

C. Parallel Simulation

A primary goal of AutoBio+ Simulation is to enable parallel data generation and evaluation for modern robot-learning methods. To achieve this, the system is designed with parallelism as a core principle, allowing rendering, visual effects, and environment simulation to run efficiently across many instances.

The Gaussian rendering pipeline supports batched rendering, allowing multiple environments to be processed simultaneously on the GPU. This vectorized design significantly improves throughput compared with sequential rendering. The particle-based visual effects are also naturally parallelizable because individual particles can be updated independently without global synchronization.

We further support texture variability by extending AutoBio’s interactive-texture framework. Because Warp does not support arbitrary texture changes at runtime, we predefine a set of textures in the MJCF scene description and index them during simulation for efficient swapping. This design extends AutoBio’s support for interactive textures while remaining compatible with parallel execution and interactive instruments in the scene. Together, these components form a unified parallel simulation pipeline capable of scaling to a large number of environments. In a representative benchmark using 339 frames from two cameras at a resolution of 640×480 , plain MuJoCo Warp rendering required 13.95 s, Gaussian-augmented rendering required 15.57 s, and Blender

rendering used by AutoBio required 1092.02 s. Thus, AutoBio+ rendering is approximately $70.14\times$ as fast as AutoBio while incurring approximately 11.61% overhead compared with plain MuJoCo Warp. This overhead represents an acceptable tradeoff given the visual benefits and the substantial speed advantage over AutoBio workflows.

IV. AUTOBIO+ BENCHMARK

A. Design Principles for Tasks

The top panel of Fig. 2 summarizes the task suite and the simulation features that support interactive assets. The AutoBio+ task design follows four principles intended to represent the manipulation capabilities required in biological laboratories rather than generic object rearrangement. (1) **Functional Tool Use.** Tasks require a policy to exploit the intended function of a laboratory tool rather than merely grasping and relocating it. (2) **Precision Manipulation.** Successful task execution requires precise manipulation and coordinated finger motion during contact-rich interactions. (3) **Language-Conditioned Visual Grounding.** A policy must associate language instructions with spatial entities in the current observation and act on the visually identified target. For example, a robot may be required to insert a tube into a rack at a location specified in the instruction, such as “row 2, column 3”. This requires grounding language descriptions into precise spatial targets and jointly reasoning over language and scene geometry. (4) **Instrument Operation.** Tasks involve interaction with laboratory instruments through articulated controls and physical manipulation, requiring the policy to operate devices as part of the task execution. These scenarios are commonly encountered in biological laboratories.

B. Dataset Generation

Grasp Synthesis. Before trajectory synthesis, we generate task-specific grasp candidates from the object geometry and the kinematic structure of the hand. Learning-based methods can directly generate grasp poses [23], [24], but require training on grasp datasets. Hence, we use Dexonomy [25] and Lightning Grasp [26] to generate a wider variety of grasp types. The resulting hand poses are represented in the object frame and converted into embodiment-specific pre-grasp and grasp targets for subsequent trajectory planning.

We instantiate the generated grasp candidates in the parallel physics simulator and evaluate them by replaying the transition from a pre-grasp configuration to the final grasp. We filter out candidates that are not reachable by the manipulator, violate collision constraints with the environment, or fail to maintain stable object contact under subsequent motion. In particular, we reject grasps that lie outside the robot workspace, induce hand–environment collisions during execution, or lead to object slippage when the arm moves. This stage ensures that the retained candidates are both kinematically feasible and dynamically stable under the contact dynamics used for demonstration collection.

Finally, we incorporate a human validation stage to refine the remaining candidates further. While some grasps are

¹<https://www.xgrids.com/intl/lixelk2>

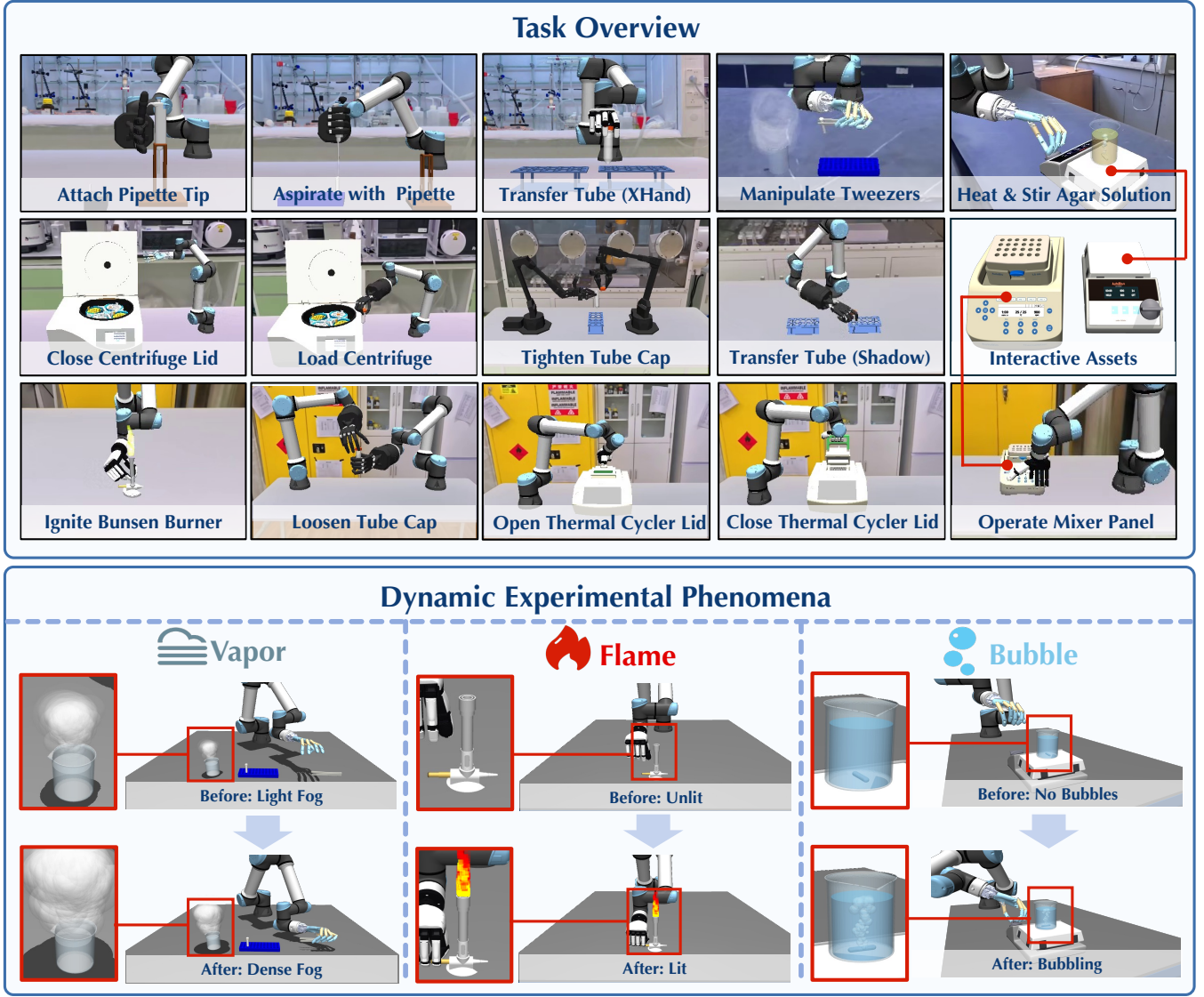


Fig. 2: **Tasks Overview and Dynamic Experimental Phenomena.** The top figure illustrates all 14 tasks in AutoBio+ along with the interactive assets, while the bottom figure presents the 3 types of dynamic experimental phenomena in AutoBio+, highlighting their visual evolution and temporal variations.

physically feasible, they may exhibit unnatural or implausible hand configurations, such as self-intersecting fingers or awkward poses that deviate from typical human grasping strategies and hinder subsequent task execution. We therefore perform manual inspection to filter out such cases, retaining grasps that are both physically valid and human-like. This step improves the quality and realism of the resulting demonstrations by aligning them with human manipulation priors. Fig. 3 (left) illustrates the overall grasp synthesis pipeline.

PFSM Policy for Data Collection. We generate demonstrations using task-specific scripted policies implemented as parallel finite-state machine. Each task is decomposed into a sequence of semantic stages (e.g., approaching and grasping), where each stage is represented by a set of end-effector waypoints and corresponding hand or gripper joint configurations. Across N parallel environments, each environment independently maintains its current stage index

q_i^t and waypoint index k_i^t . Given observation $\mathbf{o}_i^t$, the corresponding controller produces

$$\mathbf{a}_i^t = \pi_{q_i^t}(\mathbf{o}_i^t, k_i^t), \quad \mathbf{A}^t = [\mathbf{a}_1^t, \dots, \mathbf{a}_N^t]. \quad (3)$$

After each simulation step, the local state is updated as

$$(q_i^{t+1}, k_i^{t+1}) = \begin{cases} F_{q_i^t}(q_i^t, k_i^t, \mathbf{o}_i^{t+1}), & c_{q_i^t}(\mathbf{o}_i^{t+1}, k_i^t) = 1, \\ (q_i^t, k_i^t), & \text{otherwise,} \end{cases} \quad (4)$$

where c_q determines whether the current waypoint or stage is complete, and F_q advances the corresponding state. Since this update is performed independently for each environment, different task stages can be executed asynchronously within the same batched simulation. The PFSM policy data collection pipeline is illustrated on the right of Fig. 3.

To further improve throughput, we decouple physics rollout and image rendering. During rollout, camera rendering is disabled, and each environment stores its state, action,

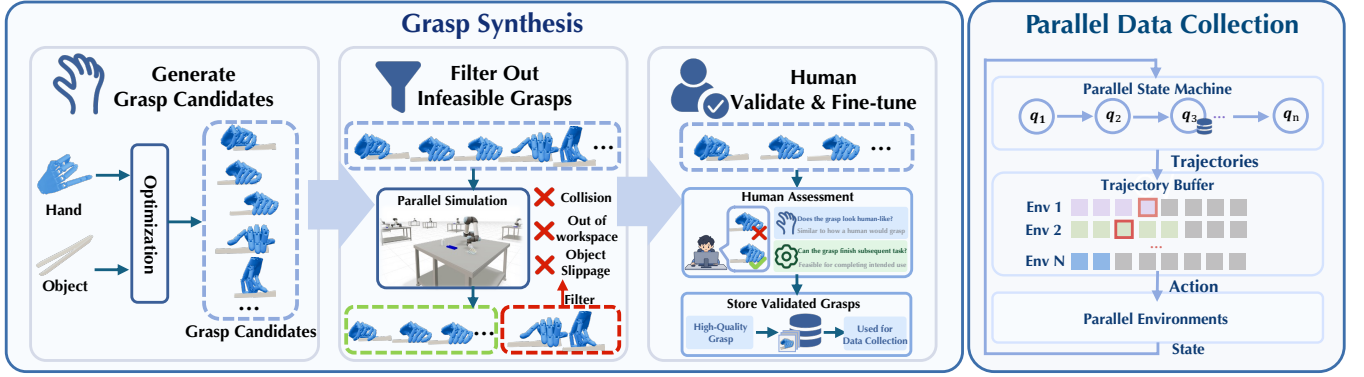


Fig. 3: **Grasp Synthesis.** First, large-scale grasp candidates are generated via optimization-based methods (e.g., Dexonomy and Lightning Grasp). Then, parallel simulation rollout is performed, and candidates are filtered based on feasibility constraints such as workspace limits and collisions. Finally, manual verification is conducted to ensure human-like plausibility. The validated grasps are stored in a grasp dataset. **PFSM Policy:** A PFSM operates over all environments in a batched manner, where each environment corresponds to an individual state instance. Transitions are triggered based on observed states, and planned trajectories are stored in a buffer that outputs batched actions when required.

timestamp, and language goal in an independent buffer D_i . Only successful trajectories are retained and replayed in batches to render multi-view RGB observations and then packaged into datasets. This procedure avoids rendering during trajectory search while preserving frame-level alignment between control and visual observations.

V. EXPERIMENTS

A. Simulation Benchmark Evaluation

We evaluate four representative policies—Diffusion Policy, $\pi_{0.5}$, GR00T-N1.7, and Fast-WAM—on 12 representative tasks selected from the full 14-task AutoBio+ suite. The selection is stratified across the four capability categories and includes all four robot embodiments. All models are trained on the same task-specific dataset and evaluated in the AutoBio+ simulator, using identical domain-randomization settings, and task-specific termination criteria. For each selected task, we report the mean success rate and SEM across 3 seeds (300 rollouts in total, with 100 rollouts per seed). All tasks use binary rewards except Operate Mixer Panel, which uses a progress reward. The resulting task-level comparison is reported in Table II and organized according to the task categories introduced in Sec. IV-A.

B. Real-World Validation

To examine whether AutoBio+ provides observations that are aligned with real-world data and meaningful for evaluating manipulation policies, we conduct a real-world validation using a UR5e robot arm equipped with an XHand and a camera system, consisting of an Intel RealSense D405 mounted on the wrist and an Intel RealSense D435 providing a third-person view, as shown in Fig. 4.

We use a few-shot data mixture to examine whether the benchmark’s simulated data capture real-world-relevant visual cues. Each training set contains 10 real-robot demonstrations and 90 simulated demonstrations. The two conditions differ only in the simulated visual background: the *Gaussian-background* condition renders the simulated demonstrations

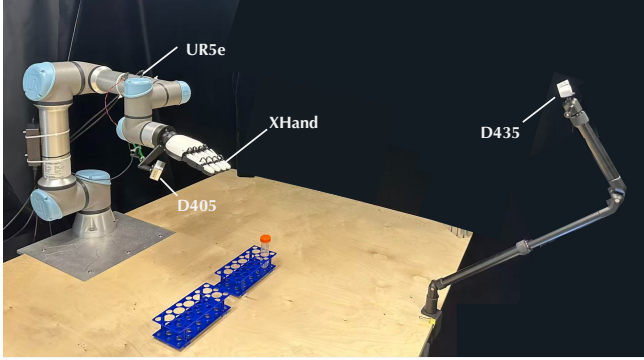
with Gaussian-splatting backgrounds, whereas the *MuJoCo-Background* condition uses the standard mlab background. We evaluate both conditions on the same real-robot task using a progressive score. The scoring rubric is: 3 points for grasping the tube, 5 points for successfully picking it up, 8 points for initiating insertion, and 10 points for fully inserting it into the rack. The tube is placed at each of ten positions in the initial rack, and each position is tested twice, resulting in 20 trials per condition. As summarized in Fig. 5, the *Gaussian-background* condition achieves a normalized mean score of 66%, compared with 17% without Gaussian backgrounds. This gap indicates that realistic background appearance can be particularly important in the few-shot regime, where the limited number of real demonstrations cannot by themselves characterize the full visual variation of the task. These results support the fidelity and practical relevance of the benchmark’s Gaussian-background data for real-world policy evaluation.

VI. DISCUSSION

Language-conditioned visual grounding remains a major bottleneck. Across language-conditioned visual grounding tasks, success rates remain generally low and highly variable across policies, and none of the tasks is solved reliably. Some models even achieve 0% success on certain tasks. These tasks require policies to ground language-specified spatial or symbolic targets in the visual scene, such as selecting a tube slot by row and column or identifying a target value on an instrument panel. The results suggest that current policies still struggle to combine language-based target identification with spatial reasoning and precise motion control. Notably, Fast-WAM does not consistently outperform the VLA baselines on these grounding tasks, suggesting that the main limitation may lie less in predicting future visual states and more in the language-conditioned front end. Improving language grounding and instruction-conditioned scene representations may therefore be necessary before the additional world-modeling capacity of WAMs can translate into better performance.

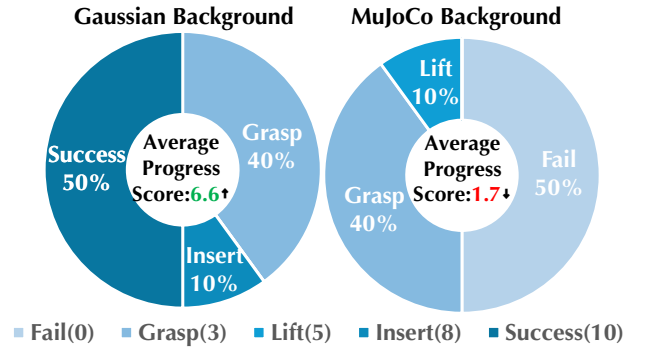
TABLE II: Task-wise success rates (%) on the 12 representative AutoBio+ tasks: mean $\pm$ SEM over 3 seeds, 100 rollouts per seed, 300 rollouts total.

Task	DP	$\pi_{0.5}$	GR00T-N1.7	Fast-WAM
<i>Functional Tool Use</i>				
Aspirate with Pipette (Shadow)	0.3 $\pm$ 0.3	0.0 $\pm$ 0.0	1.0 $\pm$ 0.0	4.7 $\pm$ 1.5
Manipulate Tweezers (Wuji)	0.7 $\pm$ 0.3	6.7 $\pm$ 2.0	4.0 $\pm$ 1.7	1.0 $\pm$ 0.6
<i>Precision Manipulation</i>				
Tighten Tube Cap (Aloha)	0.0 $\pm$ 0.0	0.7 $\pm$ 0.7	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Attach Pipette Tip (Shadow)	5.0 $\pm$ 1.2	12.0 $\pm$ 0.0	2.7 $\pm$ 0.7	20.7 $\pm$ 1.8
<i>Language-Conditioned Visual Grounding</i>				
Load Centrifuge (Shadow)	0.0 $\pm$ 0.0	1.0 $\pm$ 0.6	0.3 $\pm$ 0.3	0.7 $\pm$ 0.3
Transfer Tube (Shadow)	2.7 $\pm$ 0.9	1.3 $\pm$ 0.3	12.0 $\pm$ 1.5	0.0 $\pm$ 0.0
Operate Mixer Panel (XHand)	0.0 $\pm$ 0.0	5.5 $\pm$ 2.0	5.7 $\pm$ 1.4	0.0 $\pm$ 0.0
<i>Instrument Operation</i>				
Ignite Bunsen Burner (XHand)	60.7 $\pm$ 5.4	77.7 $\pm$ 4.8	40.0 $\pm$ 7.4	23.3 $\pm$ 3.7
Close Centrifuge Lid (Wuji)	30.0 $\pm$ 0.6	81.7 $\pm$ 1.2	100.0 $\pm$ 0.0	43.3 $\pm$ 29.7
Heat & Stir Agar Solution (Wuji)	1.7 $\pm$ 0.7	7.3 $\pm$ 1.8	7.3 $\pm$ 2.9	2.3 $\pm$ 0.9
Close Thermal Cycler Lid (XHand)	87.0 $\pm$ 1.0	5.3 $\pm$ 0.9	49.7 $\pm$ 0.7	99.7 $\pm$ 0.3
Open Thermal Cycler Lid (XHand)	14.3 $\pm$ 2.2	56.7 $\pm$ 1.2	18.0 $\pm$ 4.0	5.7 $\pm$ 3.7

**Fig. 4: Real-world setup.** A UR5e manipulator equipped with an XHand is used together with a RealSense camera system, including a wrist-mounted D405 and a third-person D435, to provide complementary observations of the workspace during policy evaluation.

Dexterous-hand pretraining alone does not ensure robust task performance. Performance varies substantially across tasks requiring dexterous-hand control. Dexterous-hand pretraining appears beneficial for relatively simple or tolerance-rich interactions. For example, GR00T-N1.7 achieves 100% success on Close Centrifuge Lid. However, this benefit does not consistently extend to contact-rich tasks that require precise alignment, fingertip coordination, and continuous finger adjustment. GR00T-N1.7 remains weak on pipette, tweezer, and tube-cap manipulation, despite involving dexterous-hand control. These results suggest that pretraining can provide a useful initialization for basic manipulation skills, but task-specific demonstrations and contact-aware control are still necessary for reliable execution of laboratory operations.

Precise contact-rich manipulation remains a prominent failure mode. Across the evaluated tasks, visuomotor policies achieve lower success rates on contact-rich tasks, such as manipulating tweezers, tightening tube caps, attaching pipette tips, loading centrifuges, and transferring tubes, but perform better on tasks involving nonprehensile interactions, such as closing centrifuge lids and opening thermal cycler

**Fig. 5: Progressive Score Distribution.** The pie charts compare the distribution of task progress across 20 real-world trials under Gaussian and MuJoCo backgrounds, showing that Gaussian backgrounds lead to better performance and average scores.

lids. These results highlight the importance of improving contact-aware control and fine-grained visuomotor coordination for reliable laboratory manipulation.

VII. LIMITATIONS AND FUTURE WORK

Task and asset scalability. Although AutoBio+ encompasses a wide range of dexterous laboratory manipulation tasks, it does not yet provide a framework for evaluating the full scope of biological laboratory automation. Future work could employ generative approaches to synthesize assets and tasks, thereby reducing manual effort and broadening the scope and content of evaluations for biological experiments.

Rendering and dynamic-scene fidelity. While AutoBio+ renders biolab scenes using Gaussian-background compositing together with a particle-based system, it may not fully reproduce high-fidelity liquid behavior or the full range of scene interactions. Future work could explore 4D Gaussian representations or dynamic neural fields to better capture liquid dynamics and scene interactions.

VIII. CONCLUSION

This paper introduced AutoBio+, a benchmark for dexterous manipulation evaluation in realistic laboratory environments with dynamic experimental phenomena. AutoBio+ comprises 14 tasks across four robot embodiments and four capability categories, and incorporates controllable flames, vapor, and bubbles to represent process-dependent visual changes. To provide realistic visual context while preserving interactive simulation, we develop a hybrid rendering framework that combines simulator-rendered foreground objects with 3D Gaussian Splatting backgrounds reconstructed from real laboratories. We further build a parallel simulation framework on top of mjlab, introducing Parallel Finite-State Machine (PFSM) for efficient demonstration generation and policy evaluation. Extensive evaluations reveal substantial variation in policy performance across tasks and expose persistent limitations in contact-rich interaction, language-spatial grounding, and fine-grained dexterous manipulation. Real-world experiments further show that Gaussian-splatting laboratory backgrounds reduce the sim-to-real gap and improve few-shot transfer compared with standard MuJoCo backgrounds, supporting the realism and practical relevance of AutoBio+ for laboratory manipulation. We hope AutoBio+ can facilitate future research on robust dexterous manipulation and perception in laboratory environments.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] Y. Chen, Y. Geng, F. Zhong, J. Ji, J. Jiang, Z. Lu, H. Dong, and Y. Yang, “Bi-DexHands: Towards human-level bimanual dexterous manipulation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 2804–2818, 2024.
- [3] Y. Yao, Z. Xu, T. Zhang, Z. Liu, S. Li, Z. Wei, F. Chen, D. Huang, K. Wan, C. Ma, S. Zhao, S. Gao, M. Tomizuka, Y. Ma, and M. Ding, “DexVerse: A modular benchmark for multi-task, multi-embodiment dexterous manipulation,” *arXiv preprint arXiv:2607.08751*, 2026.
- [4] H. Wang, W. Zhao, X. Wang, S. Huang, H. Lin, B. Zheng, R. Xu, G. Wang, Y. Mu, H. Wang, L. Fan, H. Li, Z. Zhang, and T. Tan, “DexJoCo: A benchmark and toolkit for task-oriented dexterous manipulation on MuJoCo,” *arXiv preprint arXiv:2605.16257*, 2026.
- [5] S. Li, Y. Huang, C. Guo, T. Wu, J. Zhang, L. Zhang, and W. Ding, “Chemistry3D: Robotic interaction toolkit for chemistry experiments,” in *2025 IEEE International Conference on Robotics and Automation*, 2025.
- [6] R. Li, Z. Hu, W. Qu, J. Zhang, Z. Yin, S. Zhang, X. Huang, H. Wang, T. Wang, J. Pang, W. Ouyang, L. Bai, W. Zuo, L.-Y. Duan, D. Zhou, and S. Tang, “LabUtopia: High-fidelity simulation and hierarchical benchmark for scientific embodied agents,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [7] Y. Wu, Z. Jin, T. Li, Y. Zhang, Z. Wang, S. Wang, J. Jiang, X. Li, Y. Zhang, J. Tang, Z. Che, and Y. Xia, “Labimus: A simulation and benchmark for humanoid dexterous manipulation in chemical laboratory,” *arXiv preprint arXiv:2606.31037*, 2026.
- [8] Z. Lan, Y. Jiang, R. Wang, X. Xie, R. Zhang, Y. Zhu, P. Li, T. Yang, T. Chen, H. Gao, X. Yang, X. Li, H. Zhang, Y. Mu, and P. Luo, “AutoBio: A simulation and benchmark for robotic automation in digital biology laboratory,” in *International Conference on Learning Representations*, 2026.
- [9] M. Abolhasani and E. Kumacheva, “The rise of self-driving labs in chemical and materials sciences,” *Nature Synthesis*, vol. 2, pp. 483–492, 2023.
- [10] R. D. King, J. Rowland, S. G. Oliver, M. Young, W. Aubrey, E. Byrne, M. Liakata, M. Markham, P. Pir, L. N. Soldatova, A. Sparkes, K. E. Whelan, and A. Clare, “The automation of science,” *Science*, vol. 324, no. 5923, pp. 85–89, 2009.
- [11] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Robotics: Science and Systems*, 2023.
- [12] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, vol. 44, no. 1, 2025.
- [13] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, Q. Vuong, H. Walke, H. Wang, L. Yu, and U. Zhilinsky, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025.
- [14] T. Chen, Z. Chen, B. Chen, Z. Cai, Y. Liu, Z. Li, Q. Liang, X. Lin, Y. Ge, Z. Gu, W. Deng, Y. Guo, T. Nian, X. Xie, Q. Chen, K. Su, T. Xu, G. Liu, M. Hu, H.-a. Gao, K. Wang, Z. Liang, Y. Qin, X. Yang, P. Luo, and Y. Mu, “Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation,” in *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, 2026.
- [15] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Advances in Neural Information Processing Systems*, vol. 36, pp. 44776–44791, 2023.
- [16] H. Bi, H. Tan, S. Xie, Z. Wang, S. Huang, H. Liu, R. Zhao, Y. Feng, C. Xiang, Y. Rong, H. Zhao, H. Liu, Z. Su, L. Ma, H. Su, and J. Zhu, “Motus: A unified latent action world model,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [17] T. Yuan, Z. Dong, Y. Liu, and H. Zhao, “Fast-WAM: Do world action models need test-time future imagination?,” *arXiv preprint arXiv:2603.16666*, 2026.
- [18] S. James, Z. Ma, D. Rovick Arrojo, and A. J. Davison, “RLBench: The robot learning benchmark and learning environment,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3019–3026, 2020.
- [19] C. Li, R. Zhang, J. Wong, C. Gokmen, S. Srivastava, R. Martín-Martín, C. Wang, G. Levine, W. Ai, B. Martinez, H. Yin, M. Lingelbach, M. Hwang, A. Hiranaka, S. Garlanka, A. Aydin, S. Lee, J. Sun, M. Anvari, M. Sharma, D. Bansal, S. Hunter, K.-Y. Kim, A. Lou, C. R. Matthews, I. Villa-Renteria, J. H. Tang, C. Tang, F. Xia, Y. Li, S. Savarese, H. Gweon, C. K. Liu, J. Wu, and L. Fei-Fei, “BEHAVIOR-1K: A human-centered, embodied AI benchmark with 1,000 everyday activities and realistic simulation,” *arXiv preprint arXiv:2403.09227*, 2024.
- [20] Y. Jia, G. Wang, Y. Dong, J. Wu, Y. Zeng, H. Lin, Z. Wang, H. Ge, W. Gu, K. Ding, Z. Yan, Y. Cheng, Y. Li, Z. Wang, C. Li, W. Sui, L. Shi, G. Tian, R. Huang, and G. Zhou, “DISCOVERSE: Efficient robot simulation in complex high-fidelity environments,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.
- [21] K. Zakka, Q. Liao, B. Yi, L. Le Lay, K. Sreenath, and P. Abbeel, “mjlab: A lightweight framework for GPU-accelerated robot learning,” *arXiv preprint arXiv:2601.22074*, 2026.
- [22] W. T. Reeves, “Particle systems—a technique for modeling a class of fuzzy objects,” *ACM Transactions on Graphics*, vol. 2, no. 2, pp. 91–108, 1983.
- [23] Z. Wei, Z. Xu, J. Guo, Y. Hou, C. Gao, Z. Cai, J. Luo, and L. Shao, “ $\mathcal{D}(\mathcal{R}, \mathcal{O})$ grasp: A unified representation of robot and object interaction for cross-embodiment dexterous grasping,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4982–4988, 2025.
- [24] X. Fei, Z. Xu, H. Fang, T. Zhang, and L. Shao, “ $\mathcal{T}(\mathcal{R}, \mathcal{O})$ grasp: Efficient graph diffusion of robot-object spatial transformation for cross-embodiment dexterous grasping,” *arXiv preprint arXiv:2510.12724*, 2025.
- [25] J. Chen, Y. Ke, L. Peng, and H. Wang, “Dexonomy: Synthesizing all dexterous grasp types in a grasp taxonomy,” *Robotics: Science and Systems*, 2025.
- [26] Z.-H. Yin and P. Abbeel, “Lightning grasp: High performance procedural grasp synthesis with contact fields,” *arXiv preprint arXiv:2511.07418*, 2025.